

Project Proposal: Mapping Decoy Influence in Interface Design

Summary

I'd like to extend the decoy-mapping approach of 'A map of decoy influence in human multialternative choice' (Dumbalska et al. (2020)) into interface design. Ideally, I would implement the seminar's normalisation models, and figure out where to place a decoy option on a pricing page to push the most buyers toward a certain option/pricing plan, and (as a stretch) test the predictions in an online experiment.

Motivation

The lecture's main point is that we judge value through comparison, and our brains do this by normalising to fit the situation. This has a direct applied counterpart that we see all the time. Many websites run decoy experiments on their users constantly: three-tier subscription pages, "best value" or "most popular" options, and comparison tables. The HCI research on dark patterns and choice architecture mentions these designs, but as I understand it, mostly in a descriptive way (like in Mathur et al., 2019). From what I've seen, it doesn't really use choice models to predict or quantify how much a given layout shifts decisions. The methods used by Dumbalska et al. (2020) could potentially close some of the gap by mapping decoy influence across attribute space and arbitrating between candidate models. This project applies that mapping logic to an interface context.

Research question

Where in attribute space does a decoy option most strongly change human choices in a realistic interface? Do the normalisation models that explain human decoy behaviour the best predict a different decoy influence map than less advanced models (EV/EU, range or vanilla divisive normalisation)?

Approach

Step 1: Modelling + simulation. Draw the model ladder from Seminar 2: expected value, expected utility, prospect theory, and the four normalisation types (range normalisation, divisive normalisation, recurrent divisive normalisation, and adaptive gain). Define a 2-attribute interface choice space (e.g. price vs. quality for subscription tiers, that could echo the economy/quality

space covered in Dumbalska et al., 2020). For a pair of parameters, check decoy positions across a 10x10 grid of attribute space and simulate preference shifts under each model.

Step 2: Decoy influence maps + model comparison. I would create a predicted decoy-influence map for each model. Then, I would compare maps against the published human maps generated in the property choice experiments, asking: which models create attraction, repulsion, similarity, and compromise? Then, run a sensitivity analysis over model parameters. Create likelihood-based fitting and use AIC/BIC (or something you think may be better?) for Step 3.

Step 3: Online experiment (is this feasible?). Study on Prolific (is this the right tool?): participants choose between mocked-up subscription tiers, with a decoy present and absent, and compare it to the theoretical models from previous steps

Deliverables

GitHub repo with documentation, as well as a written case study with figures, such as simulated pricing pages, decoy-influence heatmaps, and (if we do step 3), fits to data. I'd write it up to be a portfolio case study for my Master's applications.

How does it link to my essay?

My essay (Do economic choices reflect an accurate assessment of probability and value?) will argue that choices cannot accurately report value because value is constructed relative to context. The link is that the essay will hypothesise, and the project will bring it to life. I'd use the same normalisation models, but implemented with and pointed at a designed choice context to make it more relevant to me.

How can you help me?

Any input possible, a check on my framing and anything to do with the models, a pointer to anything really important that I'm missing in any literature, and, if you have advice on how to implement stage 3, it would be very welcome. I wouldn't expect more than a couple hours of your time

When?

I'd love to get this done during the course. If you think it's too time consuming, I'd be happy to extend into September.